

Re-Engineering the Connectome: Google DeepMind's Fruit Fly Brain Mapping, Embodied Physics Simulation, and the Repurposing of Insect Neural Circuits for Complex Tasks

Recent advances in automated transmission electron microscopy, machine vision, and distributed physics engines have shifted connectomics from an observational anatomical discipline into a functional branch of computational neuroscience and embodied artificial intelligence¹. By reconstructing the complete nervous system of the adult fruit fly (*Drosophila melanogaster*) at synaptic resolution, researchers from Google DeepMind, the Howard Hughes Medical Institute (HHMI) Janelia Research Campus, Princeton University, the University of Cambridge, and an international consortium have produced the first whole-brain wiring diagrams of a complex adult animal¹.

Simultaneously, the deployment of open-source biomechanical models—most notably Google DeepMind's *flybody* package in the MuJoCo physics engine—has provided an embodied substrate capable of closing the sensorimotor loop between neural dynamics and physical actuation³. These tools have rapidly moved beyond biological validation. Researchers, machine learning engineers, and independent developers are repurposing the fruit fly's neural architecture to execute complex tasks traditionally reserved for human agency or purpose-built artificial algorithms, spanning vehicle operation, first-person video games, dexterous manipulation, and symbolic text classification⁷.

Connectomic Architecture: Synaptic Blueprints of the *Drosophila* Nervous System

The realization of whole-brain computational models required scaling connectomic reconstruction methods across orders of magnitude¹¹. For decades, the structural connectome of the nematode *Caenorhabditis elegans*—comprising 302 neurons and roughly 7,000 synapses—remained the sole complete reference, yet its compact, largely non-spiking nervous system offered limited architectural guidance for complex vision-guided locomotion or spatial navigation⁵. Initial dense reconstructions in *Drosophila* were confined to sub-volumes, culminating in the 2020 release of the "hemibrain," which captured roughly 25,000 neurons and 21 million synapses across the central brain¹.

Google Research and Google DeepMind collaborated with biological consortia to resolve two full-scale connectomes that provide the architectural foundation for subsequent simulation platforms:

The FlyWire project published the complete adult female brain (FAFB) connectome in 2024,

charting 139,255 neurons and 54.5 million chemical synapses¹³. The structural pipeline employed serial-section transmission electron microscopy, slicing the cranial cavity into thousands of ultra-thin 40-nanometer sections⁵. Google applied 3D convolutional neural networks, specifically Flood-Filling Networks (FFNs) and 3D U-Nets, to automatically track contiguous cell membranes through petabytes of volumetric image data¹. Because automated segmentation introduced split and merge errors across dense neuropil regions, the FlyWire consortium developed an open-source, web-based collaborative proofreading environment that combined computational heuristics with distributed validation by researchers and citizen scientists².

In September 2026, HHMI Janelia, Google, and the University of Cambridge published the Male Central Nervous System (MaleCNS v1.0) connectome in *Cell*⁴. Spanning 166,691 neurons and more than 125 million synapses, MaleCNS represents the first connectome to encompass both the cranial brain and the entire ventral nerve cord (VNC), the insect analog of the mammalian spinal cord¹. The dataset systematically classified roughly 11,700 distinct cell types, cataloged putative neurotransmitter profiles for individual synaptic terminals, and mapped structural sexual dimorphisms¹³. Across 7,319 matched cell types in the central brain, 114 types were dimorphic, and hundreds of neurons exhibited sex-specific arborizations that directly drive distinct social and reproductive behaviors¹⁷.

Crucially, the inclusion of the VNC established the physical anatomical pathways through which descending command neurons (DNs) modulate local motor neuron pools controlling leg joints, wing hinges, and abdominal segments¹. The resulting structural graphs were standardized into public graph databases and interfaces, including Neuroglancer, neuPrint, and Codex, allowing developers to import raw biological connectivity matrices directly into deep learning and simulation pipelines¹.

Connectome Dataset	Organism / Biological Sex	Reconstructed Neurons	Synaptic Connections	Anatomical Scope	Primary Computational Role
<i>C. elegans</i> Reference ⁵	Nematode (<i>C. elegans</i>) / Hermaphrodite	302	~7,000	Whole organism	Baseline reflex loops and fundamental motor gating
Janelia Hemibrain (2020) ¹	Fruit Fly (<i>D. melanogaster</i>) / Female	~25,000	~21,000,000	Partial central brain	Central complex navigation and

					mushroom body memory
FlyWire FAFB (2024) ⁵	Fruit Fly (<i>D. melanogaster</i>) / Female	139,255	54,500,000	Complete cranial brain & optic lobes	High-resolution vision, olfactory learning, and decision-making
MaleCNS v1.0 (2026) ¹	Fruit Fly (<i>D. melanogaster</i>) / Male	166,691	>125,000,000	Whole brain and ventral nerve cord	Complete brain-to-muscle closed-loop motor actuation

Embodied Biomechanics and the Physics of Insect Simulation

Computational neuroscience increasingly demonstrates that the functional dynamics of a nervous system cannot be understood in isolation from the mechanical body it animates⁶. Without a physical morphology operating under terrestrial and aerodynamic constraints, neural networks tasked with control tend to converge onto unbiological solutions⁶. To provide a physically grounded body for connectome models, Google DeepMind partnered with Janelia to release *flybody*, an open-source, anatomically accurate biomechanical fly model integrated into the MuJoCo (Multi-Joint dynamics with Contact) physics simulator³. MuJoCo was originally designed for macro-scale robotics, such as industrial arms, quadrupeds, and humanoids, where inertial forces dominate the equations of motion¹³. At the sub-milligram mass and millimeter scale of *Drosophila*, the governing dynamics shift fundamentally: viscous drag completely overtakes inertia, and surface adhesion governs mechanical stability¹³. To simulate these phenomena, DeepMind extended MuJoCo's C core with specialized physical operators⁶. The team engineered low-Reynolds-number phenomenological fluid solvers capable of computing the instantaneous lift, translational drag, and rotational damping forces generated by high-frequency flapping wings⁶. Additionally, MuJoCo was augmented with controllable surface adhesion actuators that reproduce the van der Waals and capillary forces exerted by the fly's distal tarsal pads (pulvilli), allowing simulated agents to generate traction on vertical walls or navigate inverted along ceilings⁶.

The physical model in *flybody* was derived from high-resolution confocal fluorescence microscopy of an adult female *Drosophila*²². Researchers chemically cleared soft tissues, stained the chitinous exoskeleton, and manually segmented 67 distinct skeletal components²². These meshes were simplified to balance kinematic fidelity with computational throughput and assembled across 66 articulated joints, providing 102 controllable degrees of freedom (DoFs)²². The resulting asset captures the mechanical range of motion of the head, thorax, wings, abdominal segments, and six multi-segmented legs²¹.

In parallel, academic consortia developed *FlyGym*, the standardized engine powering *NeuroMechFly v2* at the Swiss Federal Institute of Technology Lausanne (EPFL)¹⁹. While *flybody* focused heavily on distributed deep reinforcement learning for unified walking and aerodynamic flight, *FlyGym* established full compatibility with the Gymnasium API standard, optimizing the simulation for modular neuroengineering⁶. *FlyGym* features compound eye vision models rendering retinotopic inputs across thousands of individual ommatidia, realistic antennal mechanoreceptors for wind and tactile sensing, and customizable olfactory plumes¹⁹. The closed sensorimotor loop operates continuously through these simulation platforms⁶. At each discrete timestep, the physics engine updates the environment, exposing external sensory cues (photoreceptor excitation, odor concentration, surface contact, and ambient fluid drag) alongside proprioceptive state variables (joint angles, angular velocities, and internal actuator torques)¹⁹. This multi-dimensional observation vector is injected into the sensory afferents of the neural controller²⁵.

As electrical states propagate through the network, efferent motor neurons emit continuous control signals that dictate joint torques, positional targets, and tarsal adhesion states²⁴. The physics solver integrates these forces against aerodynamic drag and ground reaction collisions, computing the resultant kinematic acceleration of the body and advancing the environment to generate the subsequent sensory frame⁶.

Computational Feature	Google DeepMind <i>flybody</i> [cite: 3, 6, 22]	EPFL <i>FlyGym</i> / <i>NeuroMechFly v2</i> [cite: 19, 24]
Underlying Physics Engine	MuJoCo (Native C bindings via Python)	MuJoCo (Migrated from PyBullet for contact stability)
Kinematic Complexity	102 Degrees of Freedom across 66 joints	42 leg DoFs (7 per leg × 6) + head and antenna joints
Locomotion Modalities	Co-simulated aerodynamic flight and terrestrial walking	Terrestrial walking, vertical wall climbing, obstacle crossing

Fluid Mechanics	Integrated low-Reynolds phenomenological wing aerodynamics	Quasi-steady aerodynamic modules and external force scripts
Subsurface Interaction	Direct MuJoCo contact solvers with tarsal adhesion actuators	Programmatic normal/shear adhesion forces ($F \approx 40 \text{ mN}$)
Control Frameworks	Distributed RL (Acme/DMPO, Ray) and Graph Neural Models	Modular Central Pattern Generators, proprioceptive feedback, RL

Bridging Static Synapses and Dynamic Sensorimotor Control

A raw connectome is fundamentally a static graph of chemical synapses; it details anatomical connectivity but omits membrane potentials, individual channel conductances, neurotransmitter receptor densities, and dynamic synaptic efficacies⁸. Translating this static matrix into a functional, embodied controller has led to two distinct computational paradigms: Graph Neural Networks trained via reinforcement learning, and biophysical whole-brain integrate-and-fire emulation²⁵.

The first paradigm is exemplified by the Fly-connectomic Graph Model (FlyGM)²⁵. Rather than replacing the brain with a generic artificial neural network, FlyGM instantiates the exact directed connectivity matrix of the adult fly as a message-passing graph neural network²⁵. The network structure is partitioned into three biologically grounded classes following the FlyWire annotation standard: afferent neurons (V_a), which receive sensory and proprioceptive input vectors; intrinsic neurons (V_i), which form the dense recurrent processing core; and efferent neurons (V_e), which drive motor outputs²⁵.

To capture physiological properties that cannot be extracted from electron microscopy, FlyGM holds the synaptic adjacency matrix W fixed as a linear-dynamical state transition operator while assigning every individual neuron a trainable intrinsic descriptor vector η_v ²⁵. These descriptors parameterize cell-specific properties such as baseline excitability, membrane leakiness, and non-linear gain²⁵.

During execution, sensory observations are injected into afferent nodes, neural states update via synapse-weighted message passing across the biological edges, and efferent states are

decoded into continuous joint commands that actuate the *flybody* model in MuJoCo²⁵. When trained via reinforcement learning across diverse locomotion tasks—including gait initiation, steady-state walking, sharp yaw turns, and aerodynamic flight—FlyGM demonstrated significantly higher sample efficiency and lower tracking error than standard multilayer perceptrons (MLPs), random graphs, or degree-preserving rewired networks²⁵. This performance gap indicates that biological connectivity provides a powerful **structural inductive bias**, restricting the policy search space to dynamically stable, low-energy sensorimotor manifolds shaped by millions of years of evolutionary selection²⁵.

The second paradigm focuses on biophysical emulation without gradient-based policy optimization²⁶. Research teams, including work by Eon Systems, have instantiated the complete FlyWire and MaleCNS connectomes within Leaky Integrate-and-Fire (LIF) frameworks executed on GPU clusters and custom Rust runtimes such as Connectome OS and FastFly⁸.

By assigning canonical signs to chemical synapses based on mapped neurotransmitter identities—utilizing acetylcholine and glutamate for excitatory connections, and GABA and histamine for inhibitory pathways—these simulations compute electrical propagation in biological real time¹³.

When coupled to *NeuroMechFly v2* in MuJoCo, the emulated nervous system demonstrated spontaneous motor coordination²⁶. Activating sugar-sensitive gustatory afferents triggered proboscis extension and feeding sequences with over 90% behavioral accuracy relative to living flies, while tactile stimulation of antennal mechanoreceptors produced coordinated anterior grooming sweeps without requiring task-specific reinforcement training or manual behavioral coding²⁶.

Repurposing Insect Neural Architectures for Human Activities and Complex Tasks

Because the complete connectome can be formulated as an open-source, differentiable graph within frameworks such as PyTorch, JAX, and TensorFlow, researchers and open-source developers have begun testing the computational boundaries of insect circuitry⁷. By connecting insect sensory inputs to human digital environments and decoding descending motor pathways into human controls, these implementations demonstrate how biological topologies adapt to tasks fundamentally outside their evolutionary history⁷.

Virtual Game Arenas and Reinforcement Benchmarks

Interactive game engines have emerged as primary testing grounds for evaluating real-time decision-making, visual perception, and plasticity in connectome models⁸. In the *Doomfly* and *FlyDoom* projects, developers interfaced the MaleCNS v1.0 graph with the *ViZDoom* first-person shooter environment⁸. The visual output of the game engine is captured on each tick and downsampled into an insect retinotopic array, projecting pixel intensities onto 3,335 luminance-sensitive R1–R6 photoreceptor inputs and 811 color-sensitive R8 photoreceptor inputs¹⁰.

Neural signals propagate through the 166,700 neurons and 25.5 million connections of the retained MaleCNS wiring¹⁰. To enable gameplay, the researchers established an engineered

motor interface: the differential firing rate of the bilateral descending neuron pair DNp20 is decoded into left and right rotational yaw, while activation in descending neuron DNpe017 commands forward translation and weapon discharge¹⁰.

To introduce functional learning during play, nonfatal in-game damage triggers an artificial aversive current into two PPL101 dopaminergic neurons¹⁰. This dopaminergic signal gates an adapted plasticity rule operating across 4,184 existing synapses between Kenyon cells and mushroom body output neurons (MBON11), mirroring the biochemical mechanism of associative punishment in living insects¹⁰. While survival in complex 3D arenas remains constrained by the absence of high-level spatial abstractions, the loop successfully demonstrates that biological reinforcement mechanisms can drive real-time evasion in dynamic virtual environments¹⁰.

Simpler video game environments provide more bounded validation⁸. In *Fly Dino*, an 80-neuron subcircuit from the MaleCNS graph was isolated to operate the Chromium desert runner game⁸. Implementing leaky hyperbolic tangent dynamics, the biological subnetwork processes incoming obstacle distances through its sensory layer and drives a single efferent jump trigger via a lightweight 243-parameter readout trained through the Cross-Entropy Method (CEM)⁸. Circuit-silencing ablation experiments demonstrated that the timing of the jump command is causally mediated by the recurrent structure of the biological subnetwork rather than the linear decoder⁸.

Similarly, in *The Fly's Table*, developers extracted a complete mushroom body microcircuit comprising 682 olfactory/state projection neurons, 1,200 Kenyon cells, 97 MBONs, and 332 dopaminergic neurons to play heads-up blackjack⁸. By encoding game score differentials into the input projection neurons, the network adjusts its hit-or-stand policies via dopamine-gated depression of specific MBON compartments, successfully optimizing betting policies through iterative play⁸.

Embodied Vehicular Driving and Kinematic Manipulation

Moving beyond purely software-based game interfaces, the *Flyhard* project instantiated a MaleCNS model inside the CARLA autonomous driving simulator to evaluate whether insect neurocircuitry could physically operate human machinery⁸. The experimental platform constructed a virtual vehicle cockpit housing a simulated fruit fly whose appendages physically interact with a scaled steering wheel and accelerator pedals⁹.

The neural controller retained 165,122 neurons and over 25.5 million synaptic connections from MaleCNS, utilizing *FlyGym* mechanics to actuate thoracic and leg joints⁹. Rather than allowing the neural network to output direct mathematical steering commands to the CARLA physics vehicle, the vehicle's turning angle was determined exclusively through the mechanical torque exerted by the fly's forelegs gripping the physical rim of the steering wheel⁹.

Over 600 optimization updates across an engineered sensorimotor interface, the connectome-driven agent learned to steer the vehicle along requested target trajectories, achieving a final angular error of less than 4.85 degrees across 100 held-out evaluation targets⁹. To verify that control was causally embodied, the researchers performed mechanical ablation tests: disconnecting the tarsal adhesion pads on the fly's forelegs eliminated over 99%

of steering authority, demonstrating that the vehicle was actively steered through physical contact mechanics rather than passive correlation⁹.

In dynamic acrobatic domains, the *fly-swing* framework integrated the *flybody* MuJoCo model into an aerial obstacle course requiring pendulum-style web swinging⁸. The agent's decision to deploy attachment cables was wired directly to the biological giant fiber escape circuit (specifically lobula columnar neurons LC4 and LPLC2 synapsing onto descending giant fiber neuron DNp01)⁸. When looming geometric obstacles exceeded critical retinal expansion rates, the giant fiber pathway fired, initiating high-velocity evasive swings that navigated complex vertical environments⁸.

Symbolic Processing and Inbox Triage

Researchers and machine learning hobbyists have also investigated whether the dense recurrent topologies of the connectome can be adapted to abstract, symbolic human tasks such as text categorization⁷. In an email classification experiment, the full *Drosophila* connectome was repurposed as a biological reservoir computing substrate⁷.

A dataset of 864 corporate emails across six operational categories—including sponsorships, consulting requests, scheduling logistics, and receipts—was converted into spatio-temporal input currents⁷. Character sequences and token n-grams were transformed into electrical pulse trains and injected into afferent sensory neuropils⁷.

Rather than modifying the fixed synaptic wiring of the brain, researchers monitored the global activation patterns of internal neuropils—creating a digital functional readout analogous to whole-brain calcium imaging⁷. A linear decoding layer was trained via supervised reward feedback to associate specific distributed firing states with document categories, triggering pre-written API response templates⁷.

The system achieved roughly 80% of the classification accuracy of a small, conventionally trained artificial neural network⁷. However, the experiment revealed steep cognitive boundaries: attempts to expand the classification scope beyond 5 to 10 categorical boundaries resulted in complete representational collapse⁷.

Because the fruit fly connectome evolved to process spatial optical flow, chemical concentration gradients, and mechanical vibrations—rather than high-dimensional symbolic structures—its internal recurrence lacks the hierarchical semantic abstractions characteristic of transformer models⁷. The project confirmed that while biological graphs can perform non-linear feature expansion for basic classification, they possess hard functional ceilings when decoupled from their native sensorimotor domains⁷.

Implement ation Platform	Underlying Connecto mic Source	Task Domain	Input Sensory Architectu re	Efferent Motor Interface	Primary Quantitativ e Finding
FlyGM	FlyWire	Embodied	Propriocept	Decoded	Outperform

Engine [cite: 25, 31]	FAFB Whole Brain	Locomotion in <i>flybody</i>	ive joint states & external tracking features	motor commands to 102 MuJoCo DoFs	ed MLPs and rewired graphs in sample efficiency and stability ²⁵
Flyhard [cite: 8, 9]	MaleCNS v1.0 (165,122 neurons)	CARLA Autonomous Driving	Road tracking angles and vehicle velocity	Physical foreleg actuation of mechanical steering wheel	Achieved <4.85° steering hold error; >99% loss of control upon grip ablation ⁹
Doomfly / FlyDoom [cite: 8, 10]	MaleCNS v1.0 (Full graph)	<i>ViZDoom</i> Arena Combat	3,335 luminance (R1–R6) & 811 color (R8) inputs	DNp20 (Yaw turn), DNpe017 (Movement & fire)	Dopamine-modulated plasticity at KC → MBON11 synapses on damage ¹⁰
The Fly's Table [cite: 8]	MaleCNS Mushroom Body Subgraph	Strategic Heads-up Blackjack	Hand numerical scores and dealer face cards	Hit vs. Stand binary decision head	Dopaminergic depression of MBONs adapts hit/stand thresholds ⁸
Connectome Email Classifier [cite: 7, 30]	MaleCNS / FlyWire Whole Brain	Discrete Text Categorization	Character / n-gram temporal current injections	Linear readout of distributed neuropil activation	Reached ~80% accuracy of baseline ANN; failed beyond 5–10

					categories ⁷
Fly Dino [cite: 8]	MaleCNS (80-neuron subcircuit)	Chromium Endless Runner	Proximity sensors for incoming obstacles	Efferent jump actuation trigger	Leaky tanh dynamics with 243-param eter CEM-traine d readout ⁸

Robotic Control, Neuromorphic Hardware, and Neurotechnological Transfer

The functional dissection of the *Drosophila* connectome is yielding significant second-order architectural insights for mainstream robotics, neuromorphic edge hardware, and neuroprosthetic rehabilitation²⁷.

In modern legged robotics, high-degree-of-freedom locomotion is traditionally formulated as an optimization problem where centralized controllers compute continuous joint trajectories and inverse dynamics at high frequencies²⁹. High-resolution connectomic analysis of the fly's ventral nerve cord has revealed a radically different design principle: fluid, multi-joint motor control is largely governed by structured networks of **reciprocal inhibitory interneurons**²⁹. Connectome-guided optogenetic and computational studies demonstrate that rhythmic leg articulation during walking and grooming does not require explicit excitatory start commands for every muscle group²⁹. Instead, premotor inhibitory circuits act as automated "macros" that enforce alternating phase relationships between opposing agonist and antagonist muscles by strategically releasing and applying inhibition²⁹.

This biological architecture resolves a critical challenge in active neuroprosthetics and humanoid robotics²⁷. Current brain-computer interfaces (BCIs) and myoelectric prostheses struggle with the high cognitive load imposed on users when attempting to decode fine-grained joint angles across multi-segmented artificial limbs²⁹.

By embedding connectome-derived reciprocal inhibitory circuits directly into prosthetic onboard microcontrollers, systems can translate a single, low-bandwidth descending intent signal into fluid, autonomous multi-joint walking or reaching motions²⁷. The user's nervous system is freed from micro-managing joint dynamics, as local inhibitory reflexes automatically maintain mechanical coordination, ground compliance, and posture²⁹.

In the domain of autonomous edge computing, the fruit fly represents a model of extreme energetic efficiency¹³. Living *Drosophila* execute full-spectrum spatial navigation, rapid visual target tracking, and dynamic flight stabilization within a power envelope of mere microwatts¹³. Conversely, modern artificial intelligence models performing comparable visual-inertial odometry require tens to hundreds of watts across heavy GPU accelerators¹³.

Translating the sparse, event-driven topology of the fruit fly brain onto neuromorphic hardware

architectures—such as Intel Loihi, SynSense DynapCNN, or custom spiking ASICs—provides a viable blueprint for micro-aerial vehicles (MAVs)¹³. Specifically, the ring-attractor compass networks identified in the fly's central complex provide a drift-free, low-power topological solution for heading estimation and path integration¹². Simultaneously, the sparse expansion layers of the mushroom body offer a structural template for one-shot visual recognition and novelty detection operating on milliwatt-scale power budgets¹³.

Biophysical Bottlenecks and the Horizon of Mammalian Connectomics

Despite rapid empirical milestones, significant theoretical and computational bottlenecks differentiate current connectome simulations from biological organisms⁸. A structural connectome reconstructed from static electron microscopy remains an anatomical snapshot⁸. Several physiological systems essential to biological function are invisible in pure electron microscopy data¹³: First, electrical synapses (gap junctions) are exceptionally difficult to resolve consistently via automated segmentation, yet they provide high-speed synchronization across flight and escape networks⁵. Second, ambient neuromodulation—the dynamic diffusion of dopamine, serotonin, and octopamine through extracellular space—acts as a global operating system that continuously alters the functional connectivity and synaptic gain of the circuit without changing physical anatomy⁸. Third, the lack of empirical channel conductances, dendritic active properties, and metabolic constraints requires simulation frameworks like FlyGM and LIF emulators to fill parameter gaps using artificial optimization or simplified biophysical assumptions²⁵.

Moving from the fruit fly to mammalian nervous systems introduces massive scaling barriers²⁶.

While the adult *Drosophila* nervous system contains approximately 1.67×10^5 neurons and 1.25×10^8 synapses, the laboratory mouse (*Mus musculus*) brain contains roughly 7.0×10^7 neurons and 1.0×10^{11} synapses—an increase of nearly three orders of magnitude in structural complexity¹.

Simulating a whole mouse brain using multi-compartmental biophysical models (such as Hodgkin-Huxley dynamics accounting for detailed dendritic arborization) will require memory bandwidth and storage exceeding 10 petabytes, surpassing the unified memory capacity of existing supercomputing clusters³³. Scaling further to the human brain (8.6×10^{10} neurons and 1.7×10^{14} synapses) remains beyond current computational and volumetric imaging regimes¹. Consequently, mammalian connectomics currently focuses on micro-volumes, such as cubic-millimeter cortical samples, while macro-scale studies rely on diffusion magnetic resonance imaging (dMRI) tractography that operates far below synaptic resolution³³.

Biological	Estimated	Estimated	Connecto	Simulation	Compute
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Model Organism	Neuron Count	Synapse Count	mic Resolution Status	Memory Requirements	Infrastructure Paradigm
Fruit Fly (<i>Drosophila melanogaster</i>) ¹	$\approx 1.67 \times$	$> 1.25 \times$	Complete whole-brain and VNC synaptic maps	$\sim$ (Sparse tensor / LIF)	Single workstation GPU / Apple Silicon MPS ⁸
Laboratory Mouse (<i>Mus musculus</i>) ²⁶	$\approx 7.0 \times$	$\approx 1.0 \times$	Dense micro-volumes; whole-brain mapping underway	$>$ (Multi-compartment)	Exascale supercomputers / Distributed GPU clusters ³³
Human (<i>Homo sapiens</i>) ¹	$\approx 8.6 \times$	$\approx 1.7 \times$	Macro-scale tractography only (synaptic mapping out of reach)	Multi-Exabyte memory regime	Beyond current high-performance computing capabilities ³

Conclusion

The release of the FlyWire and MaleCNS connectomes, coupled with Google DeepMind's *flybody* simulation framework in MuJoCo, marks an inflection point in computational neuroscience and embodied artificial intelligence¹. By grounding 166,691 reconstructed neurons within a 102-degree-of-freedom physics model, these platforms have successfully closed the sensorimotor loop in silico¹.

The broad array of experimental applications—ranging from autonomous vehicle steering in CARLA and real-time evasion in *ViZDoom* to discrete email classification and robotic manipulation—proves that biological wiring diagrams provide a functional structural inductive bias capable of stabilizing complex dynamical control⁷.

While insect neurocircuitry does not challenge deep generative models in abstract symbolic domains, it reveals fundamental principles of biological engineering: distributed reciprocal inhibition, modular motor primitives, and microwatt sensorimotor integration⁷. As connectomic imaging and physics engines scale toward mammalian systems, the fruit fly project provides the definitive operational blueprint for transforming static biological wiring into embodied,

autonomous intelligence¹.

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